



Facets of social problem-solving as moderators of the real-time relation between social rejection and negative affect in an at-risk sample

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ABSTRACT

Social rejection predicts negative affect, and theoretical work suggests that problem-solving deficits strengthen this relation in real-time. Nevertheless, few studies have explicitly tested this relation, particularly in samples at risk for suicide. This may be particularly important as social rejection and negative affect are significant predictors of suicide. The aim of the current study was to examine whether cognitive (i.e., perceiving problems as threats) and behavioral (i.e., avoidance) facets of problem-solving deficits moderated the real-time relation between social rejection and negative affect. The sample consisted of 49 young adults with past-month suicidal ideation. Demographic information, social problem-solving deficits, as well as depressive/anxiety symptoms and stress levels were assessed at baseline. Social rejection and negative affect were assessed using ecological momentary assessment over the following 28 days. Dynamic structural equation modeling was used to assess relations among study variables. After accounting for depressive/anxiety symptoms, stress levels, sex, and age, only avoidance of problems bolstered the real-time positive relation between social rejection severity and negative affect ($b = 0.04$, 95% credibility interval [0.003, 0.072]). Individuals with suicidal ideation who possess an avoidant problem-solving style may be particularly likely to experience heightened negative affect following social rejection and may benefit from instruction in problem-solving skills.

Introduction

Negative affect, such as sadness, hopelessness, and worry, is a significant risk factor for suicidal thoughts and behaviors (STB; [Armey et al., 2018](#); [Kleiman et al., 2017](#); [Turecki et al., 2019](#)). A variety of daily stressors can increase negative affect, though social stressors, such as social rejection, may play a unique role in increasing negative affect, and thus STB risk (see [Cawley et al., 2019](#); [Liu & Miller, 2014](#); [Victor et al., 2018](#)). Existing research that has employed experimental paradigms (e.g., Cyberball; see [Hartgerink et al., 2015](#) for a review) and intensive longitudinal methods (ILM; [Rabasco & Sheehan, 2021](#); [Victor et al., 2018](#)) supports the real-time positive relation between social rejection and negative affect. While it has been theorized that social problem-solving deficits may moderate this relation (e.g., [Nezu & Nezu, 2021](#)), no studies to date have examined this relation. Knowledge of the conditions under which the relation between social rejection and

negative affect may be especially strong may be used to enhance interventions for STB (see [Nock et al., 2019](#)). The current study used ecological momentary assessment (EMA; [Shiffman et al., 2008](#)) and a rigorous analytic strategy (i.e., dynamic structural equation modeling [DSEM]; [McNeish & Hamaker, 2020](#)) to examine the real-time relation between severity of social rejection and negative affect, as well as whether social problem-solving deficits strengthened this relation.

1. Real-time relation between social rejection and negative affect

The real-time relation between social rejection and negative affect is well documented. In a review of 120 studies that have used a social rejection experimental paradigm called Cyberball (see [Williams et al., 2000](#)), [Hartgerink et al. \(2015\)](#) found that social rejection has strong effects on intrapersonal constructs, including negative affect. While

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experimental paradigms have significant strengths (e.g., greater control of contextual factors that allow for stronger conclusions about causality), they may lack ecological validity (see Parsons, 2015). Research utilizing ILM may help address some of these limitations. ILM encapsulates daily diary and EMA protocols, allowing for repeated measurements of an individual's intrapersonal and interpersonal experiences as they move through the world (see Rabasco & Sheehan, 2021 for a review). Daily diary studies often ask participants to report on their collective experiences during the past 24 h (Bolger et al., 2003). On the other hand, EMA studies prompt participants *multiple times* within a 24-h period (usually through an internet-enabled device, such as a smartphone) to provide information about their experiences in that moment or since they last completed an EMA survey (Shiffman et al., 2008). EMA may be particularly well suited for assessing proximal risk factors for STB, as risk can fluctuate significantly even from hour to hour (Kleiman et al., 2017; Rabasco & Sheehan, 2021).

Several studies using ILM have documented the relation between social rejection and negative affect. For example, in a one-week daily diary study that included 121 college women, Beekman et al. (2017) found that on days participants reported greater rejection, they also reported greater negative affect. Similarly, Livingston et al. (2020) implemented EMA with 50 sexual minority adults recruited from a college campus. They collected data six times per day over 14 days. Social rejection, in the form of discrimination, significantly predicted greater anxious and depressive affect. Similar results have been found in EMA research with other marginalized groups (e.g., American-born, Black, and Latino adults; see Brondolo et al., 2008).

These experimental and observational studies of the real-time relation between social rejection and negative affect are consistent with evolutionary theory. Specifically, Baumeister and Leary (1995) posit that social relationships enabled humanity's ancestors to survive and reproduce successfully. Therefore, a "need to belong" was passed on to subsequent generations through natural selection. Affect is theorized to play a role in regulating behavior(s) that may help create and/or maintain relationships. When an individual perceives social rejection, they subsequently experience negative affect, which may motivate them to engage in behaviors (e.g., apologize for a perceived wrong-doing, seek a new relationship, etc.) that address the rejection (Baumeister & Leary, 1995). Importantly, *how* individuals address a stressor, such as social rejection, may influence the degree to which they subsequently experience negative affect.

2. Cognitive and behavioral social problem-solving deficits as moderators

Given the positive relation between social rejection and negative affect, uncovering under what circumstances, or for whom, this association is especially strong has significant implications for intervention, particularly among individuals at risk for suicide (Nock et al., 2019). Theoretical work on social problem-solving, defined as the way people identify, uncover, or create means of addressing stressors over the course of a lifetime (D'Zurilla & Goldfried, 1971), may be a potential moderator. Based on years of empirical studies, D'Zurilla et al. (2004) updated this theoretical framework to differentiate between the cognitive and behavioral dimensions of social problem-solving abilities. Specifically, *problem orientation* is defined as the cognitions a person may have about real-life problems (e.g., whether a problem is a "challenge" versus a "threat") as well as perceptions about one's (in)ability to effectively manage them (D'Zurilla et al., 2004; Nezu & Nezu, 2021). Conversely, *problem-solving style* is defined as the behavioral steps people engage in to address problems in daily living (Nezu & Nezu, 2021). Problem-solving orientations and styles often align, and a maladaptive orientation, such as perceiving problems as unsolvable and/or a major threat to one's well-being (i.e., negative problem orientation), as well as style (i.e., procrastinating on addressing problems or overly relying on others to solve problems; avoidant style), prospectively predict negative

mental health outcomes, such as depressive/anxiety symptoms and suicide risk (Nezu & Nezu, 2021). Moreover, when individuals experience stressors, negative problem orientation and avoidant style are thought to amplify the relation between a stressor and the resulting distress because the stressor may remain unresolved and even become more severe (Nezu & Nezu, 2021). Indeed, several cross-sectional studies support the role of negative problem orientation and avoidant style as moderators of the relation between stressors and negative mental health outcomes (see Nezu & Nezu, 2021 for an overview). For instance, in a sample of 123 undergraduate students, Londahl et al. (2005) found that participants' social conflict and anxiety symptoms were more strongly related in the presence of high (versus low) negative problem orientation.

While this theoretical framework posits that cognitive and behavioral social problem-solving deficits strengthen the *everyday* relation between stressors and subsequent distress, a dearth of studies exists to explicitly test this hypothesis. To the authors' knowledge, a single ILM study has partially examined this relation. Specifically, Baker (2006) recruited 39 psychology and non-psychology students, enrolled at an English university, to participate in a daily diary study. Participants' social problem-solving abilities were assessed at baseline. Five weeks later, participants were asked to complete a paper diary entry and then rate their overall affect using the Positive and Negative Affect Schedule (PANAS; Watson & Clark, 1994) at the end of each day for the next 14 days. Additionally, participants were asked to rate, on a scale from one ("extremely undesirable") to seven ("extremely desirable"), their experience of 53 everyday events that belonged to one of six categories: interpersonal, finance, work/study, environment, lifestyle, and daily maintenance. Events rated a one to three ("slightly undesirable") were defined as "hassles" while items rated five ("slightly desirable") to seven were rated as "uplifts." Items rated four ("neutral/not applicable") were not included in either category. Consistent with hypotheses, hassles positively predicted negative affect over the 14 days. Moreover, negative problem orientation and avoidant style predicted greater hassles over the 14 days. Negative problem orientation, but not avoidant style, also predicted greater negative affect over the 14 days. However, Baker (2006) did not examine whether negative problem orientation or avoidant style moderated the relation between hassles and negative affect over time. Additionally, important participant demographic information (e.g., race, ethnicity, sexual orientation) was not reported within the study, limiting conclusions about generalizability, which is not uncommon in ILM studies, particularly those that have examined factors that influence suicide risk (see Rabasco & Sheehan, 2021).

3. Current study

As described above, extant theoretical and experimental work suggests that social rejection is positively associated with negative affect, and that social problem-solving deficits, particularly negative problem orientation and avoidant style, influence daily experiences of negative affect. Though important, this work has not yet examined these relations on a more granular (e.g., hourly) level or explored how social problem-solving deficits may moderate the relation between social rejection and negative affect. Further, studies that have employed intensive longitudinal designs, particularly those that employ samples at risk for suicide, have often not incorporated participants with marginalized identities (e.g., people of color, sexual minorities, etc.; see Rabasco & Sheehan, 2021 for a review). As social rejection, problem-solving deficits, and negative affect increase risk for suicide (Armey et al., 2018; López et al., 2020), understanding their interrelation in samples at heightened risk for suicide, including those with marginalized identities, may help improve suicide prevention efforts. Using EMA, the aim of the present study was to build upon existing research by examining the real-time relation between social rejection and negative affect as well as potential moderators of this association in a diverse sample of young adults at risk for suicide (i.e., passive suicidal ideation [SI] during the past month; Liu

et al., 2020). Greater social rejection was hypothesized to predict greater negative affect over 28 days, and negative problem orientation as well as avoidant problem-solving style were hypothesized to strengthen this relation. To provide a conservative test of study hypotheses, baseline age, depressive/anxiety symptoms, stress, and sex were included as covariates (Brown et al., 2011; Hodes & Epperson, 2019; Victor et al., 2018).

4. Methods

4.1. Participants & procedures

Participants for the current study were recruited from a large, public university in the North Atlantic region of the United States from Fall 2021 to Spring 2022 using two different strategies. The university's Institutional Review Board approved the following procedures. An ad for the study was listed on the psychology research pool website and instructed students to complete a 30-min screener that included a brief consent form explaining the screener, the possibility of participating in a 2-h laboratory visit via teleconferencing software (i.e., Zoom), and completion of 28 days of EMA surveys via smartphone. Students who completed the screener through the website, regardless of their eligibility status, were awarded a half-credit. Students who were eligible and completed the laboratory visit received two additional credits.

Flyers with a QR code linking to the eligibility screener were also placed around campus to broaden the potential pool of participants. Students recruited via flyers were not compensated for the screener but were provided \$30 for the 2-h laboratory visit. Regardless of recruitment method, all participants had the opportunity to earn up to \$50 by completing EMA surveys (see below for payment structure). Inclusion criteria for the current study included: (1) 18–24 years of age; (2) fluent in English; (3) enrollment at the university as a student; (4) possession of an Android or iOS smartphone; (5) living in state for the duration of EMA period (to ensure licensed clinicians could provide support to participants should they report high suicidal intent or suicidal behavior); and (6) current, passive SI (i.e., self-reported presence of thoughts about death/dying in the past month). Exclusion criteria were: (1) cognitive impairment that would impede the ability to understand study procedures as well as potential risks; and (2) self-reported presence of autism spectrum disorder (see Venturini & Parson, 2018).

During the laboratory visit, research staff administered a semi-structured interview assessing lifetime and current STB as well as mental health symptoms (see Constructs and Measures for details). At the end of the laboratory visit, participants received training on how to complete an EMA survey and answered one practice survey. Prior EMA research suggests that 28 days is an effective time frame for assessing factors that impact suicide risk (e.g., Coppersmith et al., 2019; Kleiman et al., 2017), with shorter time frames hindering accurate assessment, even in high-risk (i.e., inpatient) samples (see Hallensleben et al., 2019). Thus, participants completed 28 days of EMA via the ExpiWell app (<https://www.expiwell.com/>) downloaded onto their smartphone during the laboratory visit. Each day, participants received one scheduled survey notification at 10:00 a.m. and five semirandom survey notifications between the hours of 11:00 a.m. and 10:00 p.m. EMA surveys assessed two domains (i.e., social rejection and negative affect; see Construct and Measures for details). To bolster survey completion rates, a payment schedule utilizing micro-incentives (i.e., small incentives for each survey; see Van Berkel et al., 2017) was implemented with bonus payments for high compliance. Specifically, participants received \$0.25 for each EMA survey completed. Additionally, each week that a participant's survey completion rate was at or above 80%, the participant received an additional \$2. At the end of each week, participants received an email with their compliance rate and how much money they had earned up until that point.

4.2. Constructs and Measures

4.2.1. Demographic information & study eligibility

To determine study eligibility, participants completed a 30-min screener that collected socio-demographic information and included the measure used to assess for problem-solving deficits (see Methods subsection Cognitive and Behavioral Facets of Problem-Solving Deficits for additional details). This screener included several questions assessing for eligibility (e.g., possession of a smartphone), including a single item that assessed for the past-month frequency of thoughts of death/dying (i.e., passive SI; Liu et al., 2020) embedded within the Patient-Reported Outcomes Measurement Information System (PROMIS) Emotional Distress – Depression Short Form (ED-DSF; Cella et al., 2010). The SI question was based on the first item of the Columbia-Suicide Severity Rating Scale-Screener-Recent-Self Report (C-SSRS-Screener-Recent-Self Report; Posner et al., 2011; Viguera et al., 2015), with answer choices created to reflect those found on the ED-DSF. Participants who endorsed “Rarely,” “Sometimes,” “Often,” or “Always” were considered to have experienced passive SI in the past month.¹

4.2.2. Lifetime/past month suicidal thoughts and behaviors

The Baseline/Screening Version of the C-SSRS (Posner et al., 2011), a semi-structured interview, was used to characterize participant's current (i.e., past-month) and lifetime STB history. Specifically, the C-SSRS evaluates four constructs: intensity and severity of SI; suicidal behavior (aborted, interrupted, actual, preparatory behavior); and lethality of suicide attempts (Posner et al., 2011). Psychometric analyses have supported its reliability and predictive value in research and clinical contexts (Interian et al., 2018).

4.2.3. Depressive/anxiety symptoms and stress levels

The Depression and Anxiety Stress Scales (DASS-21; Antony et al., 1998; Lovibond & Lovibond, 1995), a 21-item, self-report measure, was used to assess past-week severity of depressive (e.g., “I felt that I had nothing to look forward to”) and anxiety symptoms (e.g., “I felt I was close to panic”) as well as stress (e.g., “I felt I was using a lot of nervous energy”) levels. Items are rated on a four-point scale ranging from zero (“Did not apply to me at all”) to three (“Applied to me very much, or most of the time”). A total score can be obtained by summing items and multiplying the sum by two, with greater scores reflecting greater past-week depressive/anxiety symptoms and stress levels (range: 0–126; Lovibond & Lovibond, 1995). The DASS-21 has demonstrated sound psychometric properties across clinical and non-clinical samples (Antony et al., 1998; Henry & Crawford, 2005). The total score demonstrated good reliability in the current sample ($\omega = 0.88$; see Groth-Marnat & Wright, 2016).

4.2.4. Cognitive and Behavioral Facets of Problem-solving deficits

The Social Problem-Solving Inventory-Revised: Short Form (SPSI-R: SF; D'Zurilla et al., 2002), a 25-item, self-report measure was used to assess the cognitive and behavioral processes by which individuals address everyday problems (D'Zurilla et al., 2002). It contains five subscales (negative problem orientation [NPO]; positive problem orientation [PPO]; rational orientation [RO]; impulsivity/carelessness style [ICS]; avoidant style [AS]) and a total score. All items are rated on a five-point scale from zero (“Not at all true of me”) to four (“Extremely true of me”), with subscale scores ranging from 0 to 20. Only the five-item NPO (e.g., “I feel afraid when I have an important problem to solve”) and AS (e.g., “I try to do anything I can in order to avoid

¹ A correlation between the modified C-SSRS item and the frequency of suicidal ideation item on the interview version of the C-SSRS was calculated to examine convergent validity of the former. Results ($r = 0.42$, $p = .003$) are consistent with extant psychometric research on interview versus self-report measures of suicidal ideation (see Viguera et al., 2015).

problems in my life”) subscales were used in the present study because they most directly assess the complementary cognitive (i.e., perceiving problems as threats) and behavioral (i.e., avoidance of problems) facets of problem-solving deficits (see D’Zurilla et al., 2002) under study. Additionally, the five subscales are often moderately to strongly correlated (D’Zurilla et al., 2002; Hawkins et al., 2009) which can lead to issues of multicollinearity (see Cohen et al., 2003) if included in a single model, and examining them in separate models would have increased the risk of Type I errors (Benjamini & Hochberg, 1995). The NPO and AS subscales have demonstrated good psychometric properties in clinical and non-clinical samples (D’Zurilla et al., 2002; Hawkins et al., 2009). In the current sample, NPO and AS subscale scores demonstrated adequate internal consistency ($\omega_{NPO} = 0.79$ & $\omega_{AS} = 0.79$; Groth-Marnat & Wright, 2016).

4.2.5. Social rejection severity

Daily experiences of social rejection were assessed using a measured adapted from a prior daily diary study (see Beekman et al., 2017). Severity of social rejection by other people (e.g., “Since the last assessment completed, have you felt rejected or excluded by a friend?”) was assessed using a five-point scale from zero (“Did not happen”) to four (“Occurred and extremely meaningful”). The total score (range: 0–20) consisted of the sum of five items (i.e., rejected by a friend, excluded by a group, rejected by others, how excluded by other people have you felt, and how alone have you felt), with greater scores indicating greater rejection severity. Multilevel confirmatory factor analysis (CFA; Rose-llini & Brown, 2021) with a single latent factor demonstrated good fit ($\chi^2(5) = 19.00$, $p = .002^2$; CFI = 0.99; TLI = 0.94; RMSEA = 0.026; SRMR_{within} = 0.03), and the total score demonstrated adequate internal consistency ($\omega = 0.78$; Groth-Marnat & Wright, 2016).

4.2.6. Negative affect

Momentary experiences of NA were assessed using a four-item measure derived from the Positive and Negative Affect Schedule Expanded Form (PANAS-X; Watson et al., 1994). Items asked participants to rate their current experiences of NA (e.g., “How sad are you feeling right now?”) on a scale from zero (“Not at all”) to four (“Extremely”). Total NA scores (range: 0–16) are obtained by summing responses to the four items (i.e., “sad,” “hopeless,” “guilty,” and “worried”), with greater scores indicating greater momentary NA. Multilevel CFA with a single latent factor demonstrated good fit ($\chi^2(2) = 2.12$, $p = .347$; CFI = 1.00; TLI = 1.00; RMSEA = 0.004; SRMR_{within} = 0.01) and the total score demonstrated adequate internal consistency ($\omega = 0.77$; Groth-Marnat & Wright, 2016).

4.3. Data analysis plan

Descriptive statistics and distributional properties of all study variables were examined using a combination of SPSS 28 and the psych R package (Revelle, 2021). Analyses (i.e., *t*-tests and chi-square tests) were used to examine if demographic (i.e., age, sex, sexual identity, education level, race, ethnicity, recruitment method, employment level) or clinical

(i.e., frequency of passive SI in the past month) differences existed between eligible participants who enrolled (versus not) in the current study. Similar analyses were completed to determine if the recruitment strategy was related to any of the baseline variables. The overall compliance rate for the EMA surveys was calculated, as a degree of missing data is expected in studies with intensive longitudinal designs (Bolger & Laurenceau, 2013). Moreover, linear regressions were used to determine if baseline demographic (i.e., sex, age, sexual identity, race, ethnicity, employment status) and clinical (i.e., baseline levels of depressive/anxiety symptoms and stress levels, lifetime history of self-injurious behavior, severity of problem-solving deficits, recruitment strategy) variables predicted EMA completion rates. The false discovery rate was set to 5% for these analyses (Benjamini & Hochberg, 1995).

A single DSEM in Mplus 8.8 (Muthén & Muthén, 1998–2017) was used to examine whether problem-solving deficits moderated the real-time relation between social rejection severity and negative affect (see Fig. 1). DSEM utilizes Bayesian Markov Chain Monte Carlo (MCMC) estimation, which helps address convergence issues often encountered with frequentist approaches (e.g., maximum likelihood) and provides robust estimates in modest (e.g., $N = 50$) sample sizes (Asparouhov et al., 2018; Muthén & Asparouhov, 2012; Schultzberg & Muthén, 2018). Given the dearth of existing research on the current model, uninformative priors, the Mplus default, were used in current study (see McNeish & Hamaker, 2020). As a result of utilizing Bayesian methods, exact, two-tailed *p*-values are not calculated for model estimates. Rather, a credibility interval (CI) is calculated for each parameter estimate, and 95% CIs that do not include zero indicate that the corresponding estimate is statistically significant (McNeish & Hamaker, 2020). Model convergence was determined using the potential scale reduction factor (PSR), with values close to one indicating convergence. Consistent with existing research (McNeish & Hamaker, 2020; Victor et al., 2018), model estimates were first calculated using 2000 iterations of the MCMC algorithm and then calculated again with 5000 iterations to ensure that the PSR did not increase between computations.

In the current study, the hypothesized model included a mix of constructs that were measured in real time (i.e., negative affect, social rejection severity) and solely at baseline (i.e., age, sex, depressive/anxiety symptoms, stress levels, and problem-solving deficits; see Fig. 1). To create the hypothesized model, within- and between-person negative affect was regressed on within- and between-person social rejection severity, respectively. The relation between social rejection and negative affect was modeled as a latent, random slope. Then, this latent slope as well as between-person levels of negative affect were regressed on between-level predictors (i.e., age, sex, depressive/anxiety symptoms, and problem-solving deficits). To address missingness on between-level variables, their variances were estimated in the model. Between-level variables were grand-mean centered for ease of interpretation (McNeish & Hamaker, 2020). Given the wording of the social rejection items (“Thinking about the time *since the last assessment* ...”) and negative affect items (e.g., “How sad are you feeling *right now*”), and that daily social interactions may have brief temporal effects, only lagged autocorrelated residuals, and not autoregressive effects, were included in the model (see Bolger & Laurenceau, 2013; Coppersmith et al., 2019). Time was centered on the first day of EMA surveys.

5. Results

From Fall 2021 to Spring 2022, 393 students completed the eligibility screener. Of those students, 100 met eligibility criteria. Fifty of those 100 individuals completed the 2-h laboratory visit, with one participant declining to complete any EMA surveys for the current study.

² The chi-square test of fit, when significant, suggests that the model may be misspecified. However, this statistic can be oversensitive to minor model misspecifications (Kline, 2016). If model misspecifications are minor, then other indices of fit can be used to evaluate fit more accurately (Kline, 2016). Post-hoc analyses with use of modification indices (i.e., $MI > 10$; MacCallum et al., 1992), suggested three changes in the model at the within-person level (e.g., correlating residuals for two items). Changes in CFI (i.e., $\Delta < 0.01$) and RMSEA (i.e., $\Delta < 0.015$) between the original model and the less constrained models indicated that the misspecifications in the original model were minor (see Chen, 2007). Furthermore, all other fit indices (i.e., CFI, TLI, RMSEA, SRMR) for the original model indicated good fit (Hu & Bentler, 1999). Thus, the original model can be described as having good fit.

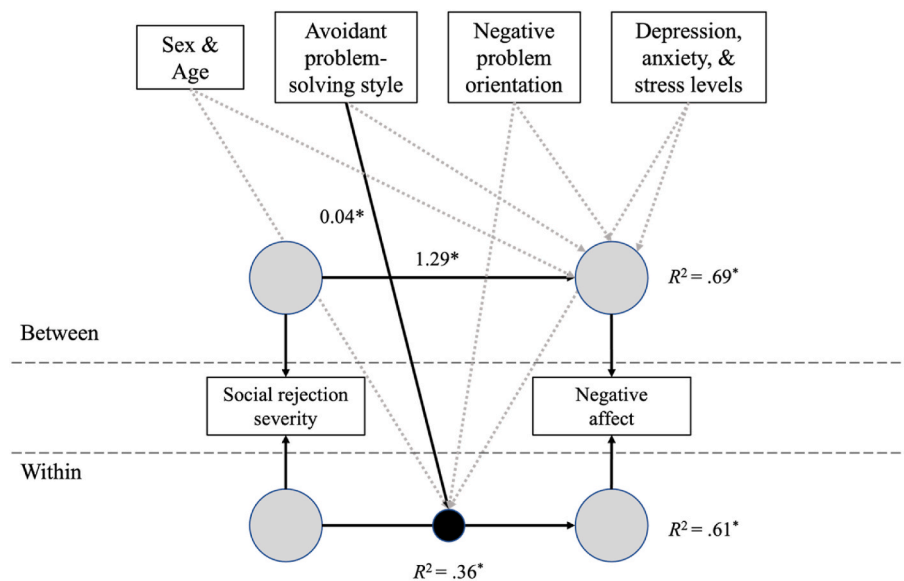


Fig. 1. Hypothesized model of the real-time relations between social rejection and negative affect and facets of social problem-solving as moderators.

Note. Social rejection severity and negative affect were measured in real-time, while all other constructs were assessed during the baseline appointment. Variance for social rejection and negative affect were partitioned into between- and within-person components and modeled as latent constructs, which are represented by gray circles in this figure. The within-person relation between social rejection and negative affect was modeled as a random latent slope, represented as a black dot in this figure. Residuals and means for latent variables as well as autocorrelations between residuals for constructs assessed in real-time were included in analyses but not represented here for ease of interpretation. Solid black arrows from observed, between-person constructs (i.e., sex, age, avoidant problem-solving style, negative problem orientation, and depression, anxiety, and stress levels) to latent constructs represent path estimates with credibility intervals that do not include zero. Gray dotted lines from observed, between-person constructs to latent constructs present path estimates with credibility intervals that include zero. Unstandardized path estimates are presented. * = estimates with credibility intervals that do not

include zero (i.e., statistically significant).

Therefore, the final sample size was 49.³ After accounting for multiple comparisons, no significant differences were found on demographic factors or frequency of past-month SI between eligible students who enrolled (versus not) in the current study (p 's > 0.05). Table 1 delineates participant demographics and descriptive statistics for study variables. Recruitment strategy was unrelated to baseline variables in the final sample for the current study (p 's > 0.05).

Missingness was present on baseline and EMA data to varying degrees. Specifically, one participant did not complete the avoidant problem-solving style subscale. Thus, the variance for this variable was declared in the DSEM to ensure all 49 participants were included in analyses. Consistent with existing EMA studies that included participants at risk for suicide (see Kivelä et al., 2022 for a review), participants overall completed approximately 51% ($M_{\text{completed}} = 86$; $SD = 53.76$; range = 5–166) of the 168 surveys over the 28-day period. No clinical or demographic variables were associated with EMA completion rates (p 's > 0.05). Recruitment strategy was also unrelated to EMA completion rates (p > .05).

As described previously, parameter estimates for the hypothesized model (see Fig. 1) were calculated twice with different numbers of MCMC algorithm iterations (i.e., 2000 and 5000) to determine if models converged appropriately. The PSR value declined from 1.07 to 1.05 between models, suggesting satisfactory convergence. Path estimates are presented in Table 2. At the between-level, social rejection severity predicted negative affect ($b = 1.30$, 95% CI [0.908, 1.695]). No other variable included in the model (i.e., avoidant problem-solving style, negative problem orientation, depressive/anxiety symptoms, stress levels, age, sex) predicted negative affect levels between participants. Overall, the model accounted for 61% of the variance in negative affect between participants ($R^2 = 0.61$, 95% CI [0.565, 0.630]), a large effect size (Cohen et al., 2003).

At the within-level, a differing pattern of results was found. Similar to between-level results, social rejection severity significantly predicted negative affect ($b = 0.40$, 95% CI [0.270, 0.543]) within a participant.

However, this within-level relation was also moderated by avoidant style, such that greater avoidant style predicted greater negative affect following more severe social rejection ($b = 0.04$, 95% CI [0.003, 0.072]). Simple slopes analyses revealed that the more a participant reported engaging in avoidant style at baseline, the greater the negative affect associated with social rejection over the 28-day period (see Fig. 2). No other variable in the model moderated this within-person relation. Predictors accounted for 36% of the variance in the relation between social rejection and negative affect ($R^2 = 0.36$, 95% CI [0.08, 0.67]), a large effect size (Cohen et al., 2003). Overall, the model accounted for 69% of the variance in negative affect at the within-person level ($R^2 = 0.69$, 95% CI [0.486, 0.824]), a large effect size (Cohen et al., 2003).

6. Discussion

The relation between social rejection and negative affect is well-documented. Nevertheless, moderators of this relation have seldom been studied using real-time designs. The present study examined whether negative problem orientation and avoidant style, cognitive and behavioral facets of social problem-solving deficits, respectively, bolstered the real-time relation between social rejection and negative affect. This question was studied in a diverse sample of at risk, young adults with past-month SI. Broadly, evidence was found for the impact of social problem-solving abilities on the real-time relation between social rejection and negative affect.

Consistent with hypotheses, avoidant style at the between-participant level moderated the within-participant relation between social rejection severity and negative affect after controlling for sex, age, baseline depressive/anxiety symptoms, stress levels, and negative problem orientation. Specifically, the more a participant endorsed characteristics of an avoidant style (e.g., avoiding a problem to “solve it”), the greater the negative affect they reported following social rejection. Though this study is the first to examine this question explicitly, these results are consistent with existing empirical research that has examined the moderating role of avoidant style on everyday stressors and affect (see Baker, 2006). They are also consistent with associated theoretical frameworks (D’Zurilla & Goldfried, 1971; D’Zurilla et al., 2004) which posit that avoiding everyday problems leaves them unsolved, with the potential for them to become even more

³ Of these 49, three participants opted to withdraw prior to the end of the 28-day EMA period due to time constraints. These participants had some EMA data and thus are included in all analyses.

Table 1
Descriptive statistics for study sample and variables of interest ($N = 49$).

Continuous variables	<i>M</i> (<i>SD</i>)	<i>N</i>	Minimum	Maximum
Negative affect	1.58 (2.66) ^a	4181 ^a	0	16
Social rejection severity	0.90 (1.98) ^a	4179 ^a	0	16
Avoidant problem-solving style	7.44 (4.64)	48	0	20
Negative problem orientation	11.02 (3.94)	49	3	17
Depressive/anxiety symptoms & stress	43.63 (24.11)	49	2	104
Age	19.73 (1.63)	49	18	24
Nominal variables	<i>n</i>	% of total <i>N</i>		
Race				
White	22	45%		
Asian	11	22%		
Black/African American	6	12%		
Other	6	12%		
Biracial/Multiracial	4	8%		
Hispanic or Latino/a/x ^b				
Yes	13	27%		
No	35	72%		
Currently employed (full- or part-time)				
Yes	25	51%		
No	24	49%		
Sex				
Female	35	71%		
Male	14	29%		
Gender identity ^c				
Cisgender	42	84%		
Transgender/gender diverse	6	12%		
Sexual identity				
Heterosexual	26	53%		
LGBQ	23	47%		
Lifetime history of suicidal behavior				
Present	18	37%		
Absent	31	63%		
Recruitment strategy				
Undergraduate research pool	20	41%		
Campus flyers	29	59%		

Note. LGBQ = lesbian, gay, bisexual, queer, or questioning. ^a Means and standard deviations were calculated over all ecological momentary assessment surveys completed by the 49 participants. ^b One participant declined to provide information on whether they identified as Hispanic or Latino/a/x. ^c One participant declined to provide their gender identity.

complex and distressing. Thus, it is possible that participants in the current study who tend to avoid their problems may not attempt to address the perceived causes of the social rejection (e.g., using assertive communication skills to address the perceived rejection), resulting in significant negative affect, particularly as these experiences accumulate over time.

Contrary to hypotheses, negative problem orientation at the between-participant level did not moderate the within-participant relation between social rejection severity and negative affect after controlling for sex, age, baseline depressive/anxiety symptoms, stress levels, and avoidant style. In other words, regardless of the degree to which a participant endorsed characteristics of a negative problem orientation (e.g., perceiving a problem as “unsolvable”), they exhibited a similar relation between social rejection and negative affect. This is inconsistent with existing ILM research examining daily relations between negative problem orientation and negative affect as well as negative problem orientation and *general* stressors (see Baker, 2006). It is possible that differences in study methodology may explain discrepant results. Unlike the current study, Baker (2006) did not include any between-level demographic (i.e., age, sex) or clinical (i.e., depressive/anxiety symptoms, stress levels) covariates. More current research suggests that these variables may play important roles in experiences of

Table 2
Path estimate summary for the real-time relation between social rejection and negative affect as well as facets of social problem-solving as moderators.

Predictor Variables	Negative Affect (Between-Level)		
	β	B (SD)	95% CI
Social rejection severity	.76	1.30 (0.20)	0.908, 1.695
Avoidant problem-solving style	.02	0.01 (0.06)	−0.112, 0.135
Negative problem orientation	.26	0.16 (0.08)	−0.009, 0.328
Depressive/anxiety symptoms & stress	−.01	−0.01 (0.01)	−0.025, 0.023
Age	.04	0.09 (0.14)	−0.183, 0.354
Sex	.07	0.51 (0.53)	−0.49, 1.583
Predictor Variables	Relation Between Social Rejection Severity and Negative Affect		
	β	B (SD)	95% CI
Avoidant problem-solving style	.42	0.04 (0.02)	0.003, 0.072
Negative problem orientation	−.29	−0.03 (0.03)	−0.080, 0.024
Depressive/anxiety symptoms & stress	.11	0.01 (0.01)	−0.004, 0.010
Age	−.03	−0.01 (0.04)	−0.086, 0.072
Sex	.02	0.03 (0.18)	−0.323, 0.378

Note. Bolded cells are statistically significant at $p < .05$. SD = standard deviation of the posterior distribution. 95% CI = 95% credibility interval for unstandardized path estimate. Variance for negative affect and social rejection was partitioned into within- and between-person components using latent centering. The relation between social rejection severity and negative affect at the within-level was calculated as a random, latent slope. All continuous variables assessed at baseline (i.e., avoidant problem-solving style, negative problem orientation, depressive/anxiety symptoms & stress, and age) were grand mean centered.

moment-to-moment negative affect (see Brown et al., 2011; Hodes & Epperson, 2019; Victor et al., 2018). As depressive/anxiety symptoms are often associated with cognitive distortions (e.g., perceiving stressors as a threat; Mercan et al., 2021), it is possible that accounting for these within the same model weakened the potential moderating role of negative problem-orientation on the relation between social rejection and negative affect. Additionally, Baker (2006) used a daily diary, rather than an EMA, design which may have precluded examination of real-time relations at a more granular level. Baker (2006) also explored a combination of stressors (i.e., interpersonal, finance, work/study, environment, lifestyle, and daily maintenance) whereas the present study only examined experiences of social rejection. It is possible that problems with different degrees of emotional and social consequences (e.g., not brushing one's teeth versus breaking up with romantic partner) may be differentially impacted by negative problem orientation. Finally, it is possible that negative problem orientation may have played a *relatively less* important role than avoidant style as a moderator in the real-time relation between social rejection and negative affect. Transdiagnostic models of psychopathology highlight the role of cognitive fusion, defined as the degree to which an individual believes the literal meaning of their thoughts rather than perceiving thoughts as transient, malleable experiences (Bardeen & Fergus, 2016). Thus, while individuals may initially (and generally) perceive events as a significant threat (e.g., “I will not pass my dissertation defense”), they may not be fully convinced that the thought is accurate, ultimately approaching (e.g., defend dissertation results), rather than avoiding (e.g., never defend dissertation results), the problem to solve it. Future ILM research that incorporates measures of cognitive fusion may help determine if this is a plausible hypothesis for this pattern of results.

6.1. Clinical implications

Results from the current study hold implications for clinical practices with young adults with past-month SI. First, assessment of social problem-solving abilities prior to the start of treatment may help inform treatment planning with this group. Second, addressing behavioral avoidance of problems at the start of care may help weaken the relation between social rejection and negative affect, and thus suicide risk. Of

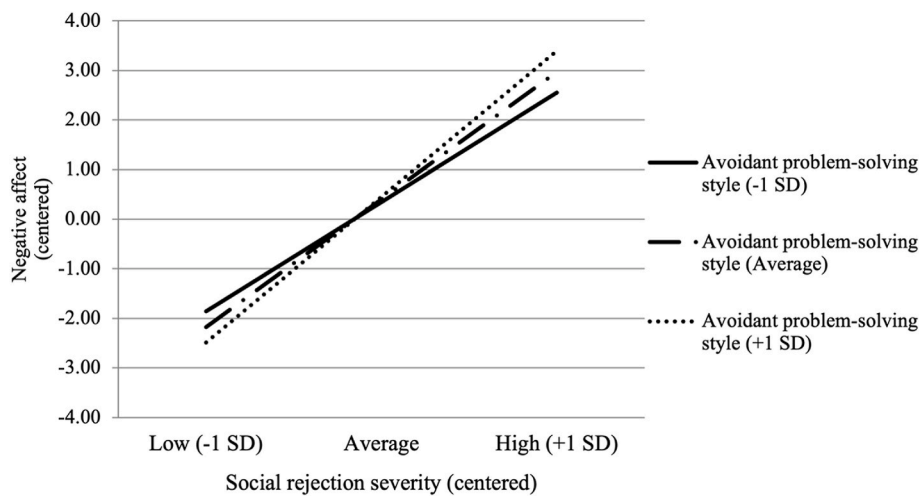


Fig. 2. Simple slopes analyses for the real-time relation between social rejection and negative affect with avoidant problem-solving style as the moderator.

Note. Predictor (i.e., social rejection severity) and outcome (i.e., negative affect) were centered using latent centering in Mplus. The moderator (i.e., avoidant problem-solving style) was grand-mean centered. Traditional alpha values (i.e., p -value) were not calculated due to use of Bayesian estimation, though credibility intervals (CI) were calculated for each simple slope. CI that do not contain zero are considered statistically significant from a frequentist perspective. The simple slopes at low (i.e., -1 SD; $b_{\text{low}} = 0.23$, 95% CI [0.020, 0.438]), average ($b_{\text{average}} = 0.40$, 95% CI [0.270, 0.543]), and high (i.e., $+1$ SD; $b_{\text{high}} = 0.58$, 95% CI [0.365, 0.780]) levels of the moderator were all statistically significant. Additionally, each simple slope was significantly different the others (i.e., $b_{\text{high}} > b_{\text{average}} > b_{\text{low}}$).

note, dismantling studies of randomized clinical trials testing problem-solving treatment protocols have found that failing to address a negative problem-orientation leads to poorer mental health outcomes post-treatment (e.g., greater depressive symptoms; see Nezu & Nezu, 2021 for an overview). Therefore, techniques from evidence-based, cognitive-behavioral treatment protocols, such as emotion-centered problem-solving therapy, can help reduce dependence on avoidance as a problem-solving strategy (e.g., problem definition and formulation; externalization; see Nezu & Nezu, 2021) as well as address negative problem orientation. Regarding social rejection in particular, these results suggest that it will be important for clinicians to help their clients find a way to address these types of experiences, even if perceived as a threat. Clinicians may work with clients to enhance understanding of how avoidance may hold short-term benefits (e.g., immediate relief from discomfort) but long-term costs (i.e., greater negative affect following social rejection). Instructing clients on the use of assertive communication skills, such as DEARMAN (Linehan, 1993, 2015), may also support clients in directly addressing some forms of social rejection (e.g., situations in which a misunderstanding between individuals may be perceived as rejection). When social rejection cannot be directly changed by an individual (e.g., being discriminated against due to sexual identity), clinicians may be able to support clients by instructing them in the use of radical acceptance (e.g., focusing on what can be changed, such as the emotional fallout, using distress tolerance skills; Linehan, 1993, 2015). Relatedly, if one problem cannot be “solved” by an individual (e.g., societal discrimination), it is possible that such experiences may shape that individual’s perception of *all* problems as unsolvable (e.g., see Metzger et al., 2020). Thus, using a collaborative approach, clinicians can support clients by jointly reflecting on how solvable a problem may be and selecting an effective course of action (e.g., thinking of all options to solve the specific problem versus use of radical acceptance to manage affective response). Last, existing experimental work suggests that experiencing social inclusion following social rejection may help reverse the detrimental effects of rejection (Tang & Richardson, 2013). Therefore, clinicians may help at-risk clients create a “cope ahead” plan (Linehan, 1993, 2015) that includes engaging with supportive peers or family following rejection to effectively reduce their negative affect.

6.2. Strengths, limitations, and future directions

This study has several key strengths, including an intensive longitudinal design, rigorous analytic strategy, and use of a diverse sample. Nonetheless, it also has limitations that necessitate mention. First, while the current study employed measures used in prior intensive longitudinal research, data was solely collected via self-report. Novel research

that has utilized passive measurement of physiological distress (e.g., electrodermal activity) in real-time suggests that such data, when combined with self-report data collected via EMA, can bolster our ability to predict, and prevent, sharp increases in suicide risk (Kleiman et al., 2021). Relatedly, the social rejection measure used in the current study provided an overall assessment of social rejection severity. Existing EMA research suggests that rejection and/or social stressors that involve specific individuals in a person’s life or degree of closeness to a person (e.g., family members versus friends; Husky et al., 2017) may play relatively more important roles in increasing suicide risk. Finally, while all participants in the current study were at increased risk for suicidal behavior (see Liu et al., 2020) and may have been in mental health treatment at the time of study participation, they were recruited from a university setting, which may limit the generalizability of results. Future research with individuals recruited from clinical settings that incorporate multimethod approaches and directly identify the sources of social rejection is necessary to overcome these limitations.

Nevertheless, this study extends existing research by assessing how cognitive and behavioral facets of social-problem deficits moderate the real-time relation between social rejection and negative affect. Results suggest that helping clients address their tendency to avoid problems may be particularly helpful in reducing negative affect following social rejection, and thus suicide risk, among young adults.

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ORCID iD authorship contribution statement

Roberto López: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Software, Project administration, Resources, Visualization, Roles, Writing – original draft. **Christianne Esposito-Smythers:** Supervision, Validation, Writing – review & editing. **Annamarie B. Defayette:** Methodology, Resources. **Katherine M. Harris:** Investigation, Project administration, Resources. **Lauren F. Seibel:** Investigation, Project administration, Resources. **Emma D. Whitmyre:** Resources.

Declaration of competing interest

All authors declare that they have no conflict of interest.

Data availability

Data will be made available on request.

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